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五种商品意大利面在碱水煮制下的尺寸变化、时间进程与中式面条适配性：一项探索性厨房比较

Dimensional Changes, Time Course, and Chinese-Noodle Applications of Five Commercial Pasta Types Cooked in Alkaline Water: An Exploratory Kitchen Comparison

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Figure 1. Finished-dish presentation documented by the author. Commercial products were independently purchased.

摘要与关键词 / Abstract and Keywords

摘要： 本探索性厨房比较在统一碱水条件下考察五种商品意大利面的尺寸变化、建议煮制时间和中式面条应用潜力。Angel Hair、Thin Spaghetti、Spaghetti、Linguine和Fettuccine均在1 L沸水加入4 g小苏打的条件下煮制，并在作者判断的“刚好弹牙”端点记录时间。所报告的生面/熟面尺寸分别为1.2/1.8、1.4/2.1、1.8/2.8、3.1/4.6和4.6/6.8 mm；对应建议时间为5、9、15、17和18分钟。五种产品的平均尺寸膨胀倍数为1.50。Angel Hair的3至8分钟时间序列显示，测量直径由1.5增加至1.9 mm，但推荐感官端点为5分钟，说明最大膨胀并不等同于最佳口感。名义生面尺寸与建议煮制时间呈描述性正相关（Pearson r = 0.82），但样本仅包含五种产品，且圆面使用直径、扁面使用宽度，因此该相关不能用于推断一般规律。结果可为家庭烹饪者选择面型和时间提供初步参考，同时也提出了标准化测量与重复实验的需要。

Abstract: This exploratory kitchen comparison examined dimensional change, recommended cooking time, and potential Chinese-noodle applications for five commercial pasta formats under a standardized alkaline cooking condition. Angel Hair, Thin Spaghetti, Spaghetti, Linguine, and Fettuccine were cooked in 1 L boiling water containing 4 g baking soda, and cooking time was recorded at the author-defined al dente endpoint. Reported dry/cooked dimensions were 1.2/1.8, 1.4/2.1, 1.8/2.8, 3.1/4.6, and 4.6/6.8 mm, with recommended times of 5, 9, 15, 17, and 18 minutes. Mean dimensional expansion was 1.50. An Angel Hair time series from 3 to 8 minutes increased from 1.5 to 1.9 mm, although 5 minutes was selected as the preferred sensory endpoint, indicating that maximum swelling did not equal optimal texture. Nominal dry size and recommended time showed a descriptive positive association (Pearson r = 0.82). However, the dataset included only five products and mixed diameter measurements for round pasta with width measurements for flat pasta. The findings offer an initial practical reference while highlighting the need for replication and standardized measurement.

关键词 / Keywords:
碱水煮制；卡尺测量；尺寸膨胀；意大利面规格；中式面条替代 / alkaline cooking; caliper measurement; dimensional expansion; pasta format; Chinese noodle substitute

1. 研究背景 / Background

不同意大利面规格在粗细、截面形状和包装建议时间上差异明显。若统一采用碱水煮制，家庭烹饪者需要知道细面是否更快达到理想端点、扁面与圆面的尺寸变化是否接近，以及不同规格更适合哪些中式面条菜式。本研究在第一篇4 g/L小苏打条件的基础上，将比较扩展到五种常见规格。

Commercial pasta formats differ in strand thickness, cross-sectional geometry, and conventional cooking instructions. Under a common alkaline cooking condition, a home cook needs to know whether finer strands reach a preferred endpoint sooner, whether flat and round pasta show similar relative dimensional change, and which formats may suit particular Chinese-style dishes. Building on the 4 g/L condition identified in the first report, this study compared five common formats.

2. 材料与方法 / Materials and Methods

样品。 比较对象为Angel Hair、Thin Spaghetti、Spaghetti、Linguine和Fettuccine。原始图片记录了商品包装和名义规格。所有产品由作者自行购买。

Samples. Angel Hair, Thin Spaghetti, Spaghetti, Linguine, and Fettuccine were compared. Product packaging and nominal dimensions were documented photographically. All products were independently purchased by the author.



Figure 2. Commercial pasta products included in the comparison.

统一条件。 每种面条在1 L沸水加入4 g小苏打的条件下煮制。达到作者判断的“刚好弹牙”状态时记录时间。该端点为感官判断，并非仪器终点。

Standardized condition. Each pasta type was cooked in 1 L boiling water containing 4 g baking soda. Time was recorded at the author-defined al dente endpoint. This endpoint was sensory rather than instrumental.

尺寸测量。 使用数显卡尺记录尺寸。三种圆形面记录直径；Linguine和Fettuccine记录宽度。由于熟面柔软且可压缩，卡尺夹持力度可能影响读数。原始材料未说明每种面条测量了多少根、是否进行了重复测量或卡尺是否校准。

Dimensional measurement. A digital caliper was used. Diameter was recorded for the three round formats, whereas width was recorded for Linguine and Fettuccine. Cooked pasta is soft and compressible, so caliper pressure could affect readings. Strand count, repeated measurement, and calibration were not documented.

派生指标。 尺寸膨胀倍数按熟面尺寸除以生面尺寸计算。五种产品的平均倍数为1.50。生面名义尺寸与建议时间的Pearson相关系数为0.82，仅用于描述。

Derived measures. Expansion ratio was calculated as cooked dimension divided by dry dimension. The mean ratio was 1.50.

Pearson correlation between nominal dry dimension and recommended time was 0.82 and is reported descriptively only.

3. 结果 / Results

Type	Dry	Cooked	Ratio	Time
Angel Hair	1.2	1.8	1.50x	5 min
Thin Spag.	1.4	2.1	1.50x	9 min
Spaghetti	1.8	2.8	1.56x	15 min
Linguine*	3.1	4.6	1.48x	17 min
Fettuccine*	4.6	6.8	1.48x	18 min

* Linguine and Fettuccine values are widths; round-pasta values are diameters.

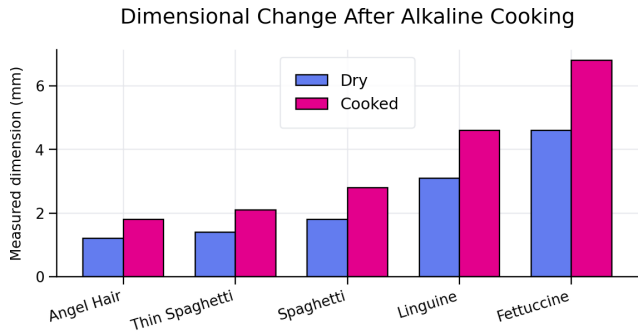


Figure 3. Reported dry and cooked dimensions.

相对膨胀倍数集中在1.48至1.56之间，说明在本次测量中不同规格的相对尺寸增长较为接近；然而绝对增长量随规格增大而增加。标准Spaghetti报告的比例最高（1.56），两种扁面约为1.48。

Relative expansion ratios clustered from 1.48 to 1.56, suggesting similar proportional dimensional growth within this small dataset, while absolute growth increased with pasta size. Standard Spaghetti had the highest reported ratio, 1.56, whereas the two flat formats were approximately 1.48.

建议煮制时间从Angel Hair的5分钟增加到Fettuccine的18分钟。名义尺寸与建议时间呈正相关，但标准Spaghetti位于简单线性趋势之上，而Fettuccine位于趋势之下，提示截面形状、配方和感官端点同样重要。

Recommended time increased from 5 minutes for Angel Hair to 18 minutes for Fettuccine. Nominal size and recommended time were positively associated, but standard Spaghetti sat above a simple linear trend and Fettuccine below it, indicating that geometry, formulation, and sensory endpoint also mattered.

4. Angel Hair时间进程 / Angel Hair Time Course

Angel Hair在3、4、5、6、7、8分钟的读数分别为1.5、1.6、1.7、1.8、1.8和1.9 mm。3分钟时仍可见硬芯；5分钟被选为推荐端点。6分钟以后尺寸继续增加，但原始观察指出面条更容易偏软。

Angel Hair readings at 3, 4, 5, 6, 7, and 8 minutes were 1.5, 1.6, 1.7, 1.8, 1.8, and 1.9 mm. A firm center remained visible at 3 minutes, and 5 minutes was selected as the recommended endpoint. Measured size continued to increase after 5 minutes, but the source observations described progressive softening beyond approximately 6 minutes.

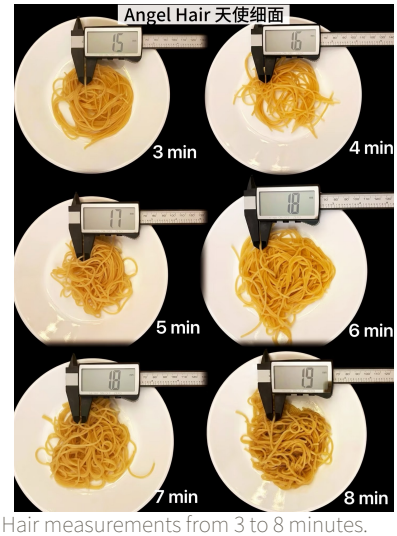


Figure 4. Angel Hair measurements from 3 to 8 minutes.

该序列说明，尺寸膨胀是持续的物理变化，而“最佳口感”是由面芯、表面软化和弹性共同决定的感官端点，两者不应混为一谈。

The series shows that dimensional swelling is a continuing physical change, whereas preferred texture is a sensory endpoint determined jointly by center doneness, surface softening, and elasticity. The two concepts should not be treated as equivalent.

5. 中式面条应用 / Chinese-Noodle Applications

Angel Hair可作为细面、凉面或馄饨面风格的起点，但吸水快、容错窗口短。Thin Spaghetti在细度和耐煮性之间较平衡，可用于阳春面或细拌面。标准Spaghetti的适用范围最广，可用于汤面、热干面风格或炒面实验。Linguine的扁平表面有利于酱汁附着，适合拌面和炒面。Fettuccine宽度较大，可用于烩面或宽面风格。以上建议描述的是烹饪适配性，不表示原料、加工工艺或物理结构与对应中式面条完全相同。

Angel Hair offers a starting point for fine-noodle, cold-noodle, or wonton-noodle-style dishes, but its rapid hydration creates a narrow timing window. Thin Spaghetti balances fineness and cooking tolerance. Standard Spaghetti appears the most versatile for soup, mixed, or stir-fried applications. Linguine’s flat surface supports sauce adhesion, while Fettuccine may suit broad-noodle-style dishes. These suggestions concern culinary use and do not imply equivalence in ingredients, processing, or structure.



Figure 5. Original five-format results and application suggestions.

6. 讨论 / Discussion

五种产品都接近1.5倍的尺寸膨胀，为家庭烹饪提供了一个易记的经验值，但该数字不能替代对单一产品的实际观察。尤其是扁面的宽度变化与圆面的直径变化并非同一物理量，不能据此推断截面积、体积或吸水率。

An approximate 1.5-fold dimensional change offers a memorable practical observation, but it should not replace direct observation of a specific product. Width change in flat pasta and diameter change in round pasta are not the same physical quantity and cannot be used to infer cross-sectional area, volume, or water uptake.

Pearson r = 0.82看似较高，但仅基于五个异质产品。没有重复测量、误差范围或预先设定的统计分析，因此不应报告显著性或建立预测公式。该图的价值主要在于显示“规格通常越大，煮制时间越长”的方向性趋势。

Although Pearson r = 0.82 appears high, it is based on only five heterogeneous products without replication, uncertainty estimates, or a prespecified statistical analysis. Significance testing and prediction equations would therefore be inappropriate. The graph is useful primarily for showing the directional tendency that larger formats generally required longer cooking.

7. 局限性 / Limitations

本研究未说明每种产品测量的面条根数、取样位置、卡尺夹持力度、重复次数和设备校准。包装名义尺寸可能来自制造商而非本次实验直接测量。感官端点由一位作者确定，且不同品牌的蛋白质含量、挤压工艺和干燥条件可能影响结果。全麦意大利面和泡打粉仅有描述性尝试，不纳入定量比较。

The strand count, sampling location, caliper pressure, number of repeats, and instrument calibration were not documented. Nominal dry dimensions may have come from packaging rather than direct measurement. One author selected the sensory endpoint, and differences in protein content, extrusion, and drying may influence results. Descriptive trials involving whole-wheat pasta and baking powder were not included in the quantitative comparison.

8. 结论 / Conclusion

在统一4 g/L小苏打条件下，五种规格的建议时间为5至18分钟，平均尺寸膨胀约1.50倍。较大规格一般需要更长时间，但几何形状和感官端点同样重要。后续研究应使用重复测量、固定卡尺压力、记录品牌与配方，并结合烹煮损失和仪器质构。

Under a standardized 4 g/L baking-soda condition, recommended cooking times ranged from 5 to 18 minutes and mean dimensional expansion was approximately 1.50. Larger formats generally required longer cooking, while geometry and sensory endpoint also mattered. Future work should use repeated measurements, controlled caliper pressure, product-composition records, cooking-loss analysis, and instrumental texture testing.

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